ALIZA FARAZ

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TECHNICAL SKILLS

Languages: Java, C, C++, Python, JavaScript, HTML5, CSS3, Lua
Frameworks & Libraries: p5.js, ml5.js (PoseNet), REST APIs
Tools & Platforms: KiCad, Git/GitHub, VS Code, Arduino IDE, Roblox Studio
Technical Concepts: Object-Oriented Programming (OOP), Hardware-Software Integration, Embedded Firmware, Serial Communication, Agile Development

PERSONAL PROJECTS

Custom Embedded Keyboard — Hardware-Software Integration Project, *July 2026 – September 2026*

- Designed and routed a custom electronic printed circuit board (PCB) schematic using KiCad, optimizing trace layout, signal integrity, and component placement.
- Programmed microcontrollers with customized embedded firmware to handle real-time matrix scanning, debouncing logic, and input-output (I/O) mapping.

MindBridge – Mental Wellness Monitor, *May 2025 – June 2025*

- Built a hardware-software system using JavaScript and p5.js that rendered Arduino sensor inputs as a real-time browser interface via serial communication.
- Integrated PoseNet (ml5.js) to detect yoga poses as mindfulness triggers, demonstrating use of machine learning APIs in a cross-platform environment.

OurCity – Civic Engagement Web App, *July 2023*

- Engineered a web application prototype facilitating youth civic participation and city sustainability proposals in alignment with UN SDG 11.

EDUCATION

Bachelor of Engineering, Engineering Systems and Computing (Co-op), *2025 – Present*
University of Guelph, Guelph, ON

Selected Academic Project:

Automated Teddy Bear Wheelchair, *2025*

- Designed and built a functional motorized wheelchair prototype by implementing motor driver logic with Arduino code to manage directional movement; project earned a grade of approximately 90%.

EXTRACURRICULAR EXPERIENCE

Recipient, STEM Digital Citizenship Graduation Award, *June 2025*
Ontario Ministry of Education, Ontario

- Selected as one of two recipients province-wide for outstanding STEM leadership; recognized by the Lieutenant Governor of Ontario.

Founder & President, Student Hack Club, *Nov. 2024 – June 2025*
Hack Club Organization, Milton, ON

- Founded and led a school-wide coding club, organizing technical workshops on programming and software development for 20+ students.

Program Participant, hEr VOLUTION STEMing UP, *July–Aug. 2022 & 2023*
hEr VOLUTION, Toronto, ON

- Completed two years of intensive technical training building web applications aligned with UN SDGs, presenting solutions to a panel of technical evaluators.

WORK EXPERIENCE

Python Instructor & Curriculum Developer, *June 2026– August 2026*
hEr VOLUTION, Ontario

- Developed and delivered comprehensive Python programming curriculum, introducing core computational logic, data structures, and algorithmic thinking to students.